

CONTENTS

MODERN PHYSICS–2

• Key-Concepts :	Formulae and Basic Theory	1
• Exercise (S-1) :	Conceptual Subjective Problems	14
• Exercise (S-2) :	Challenging Subjective Problems	15
• Exercise (O-1) :	Miscellaneous Type Problems	17
• Exercise (O-2) :	Miscellaneous Type Problems	22
• Exercise (J-M) :	Previous 10 years AIEEE Problems	26
• Exercise (J-A) :	Previous 10 years IIT-JEE Problems	29
• Answer Key		35

The difference between the total mass of the nucleons and mass of the nucleus is called the mass defect of the nucleus represented by $\Delta m = [Zm_p + (A-Z)m_n] - m_{\text{nuc}}$

Multiplying the mass defect by the square of the velocity of light, we can find the binding energy of the nucleus.

$$BE = \Delta mc^2 = [(Zm_p + (A-Z)m_n) - m_{\text{nuc}}]c^2$$

If the masses are taken in atomic mass unit, the binding energy is given by

$$BE = [(Zm_p + (A-Z)m_n) - m_{\text{nuc}}] 931.5 \text{ MeV}$$

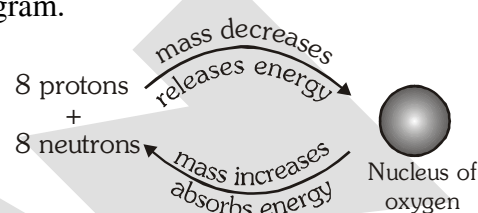
Let us take example of oxygen nucleus. It contains 8 protons and 8 neutrons.

We can discuss concept of binding energy by following diagram.

$$8m_p + 8m_n > \text{mass of nucleus of oxygen}$$

For nucleus we apply mass energy conservation,

$$8m_p + 8m_n = \text{mass of nucleus} + \frac{B.E.}{c^2}$$



For general nucleus ${}_Z^AX$, mass defect = difference between total mass of nucleons and mass of the nucleus

$$\Delta m = [Zm_p + (A-Z)m_n] - M$$

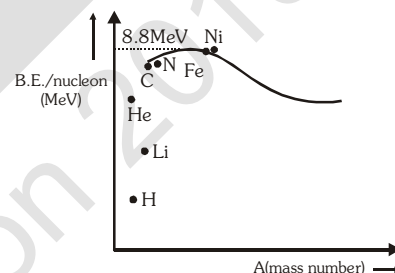
$$B.E. = \Delta mc^2 \text{ (joules)} = (\Delta m)_{\text{in amu}} \times 931.5 \text{ MeV}$$

Binding Energy per Nucleon

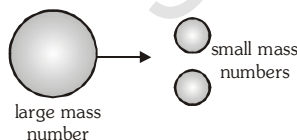
Stability of a nucleus does not depend upon binding energy of a nucleus but it depends upon binding energy per nucleon

$$B.E./\text{nucleon} = \frac{B.E.}{\text{mass number}}$$

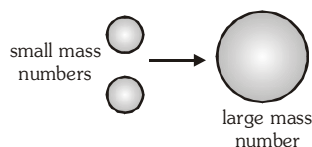
$$\text{Stability} \propto \frac{B.E.}{A}$$



- (i) $B.E./A$ is maximum for $A = 62$ (Ni), It is $8.79460 \pm 0.00003 \text{ MeV/nucleon}$, means most stable nuclei are in the region of $A = 62$.
- (ii) Heavy nuclei achieve stability by breaking into two smaller nuclei and this reaction is called fission reaction.



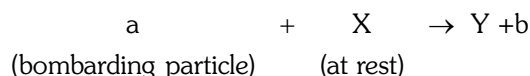
- (iii) Nuclei achieve stability by combining and resulting into heavy nucleus and this reaction is called fusion reaction.



- (iv) In both reactions products are more stable in comparison to reactants and Q value is positive.

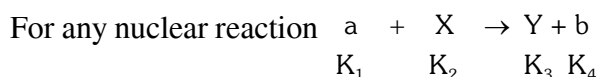
NUCLEAR COLLISIONS

We can represent a nuclear collision or reaction by the following notation, which means $X(a,b)Y$



We can apply :

- (i) Conservation of momentum (ii) Conservation of charge (iii) Conservation of mass-energy



By mass energy conservation

$$(i) K_1 + K_2 + (m_a + m_x)c^2 = K_3 + K_4 + (m_Y + m_b)c^2$$

(ii) Energy released in any nuclear reaction or collision is called Q value of the reaction

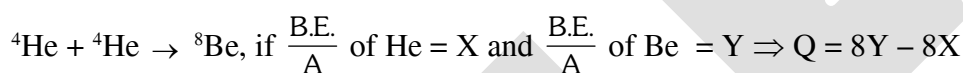
$$(iii) Q = (K_3 + K_4) - (K_1 + K_2) = \Sigma K_p - \Sigma K_r = (\Sigma m_r - \Sigma m_p)c^2$$

(iv) If Q is positive, energy is released and products are more stable in comparison to reactants.

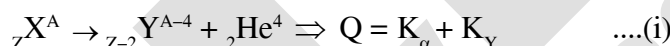
(v) If Q is negative, energy is absorbed and products are less stable in comparison to reactants.

$$Q = \Sigma(B.E.)_{\text{product}} - \Sigma(B.E.)_{\text{reactants}}$$

Ex. Let us find the Q value of fusion reaction



Q value for α decay



Momentum conservation, $p_Y = p_\alpha \quad \dots(ii)$

$$K_\alpha = \frac{p^2}{2 \times m \times 4}$$

$$K_Y = \frac{p^2}{2m(A-4)} = \frac{4K_\alpha}{A-4}$$

$$Q = K_\alpha + \frac{4K_\alpha}{A-4} = \frac{A}{A-4} K_\alpha$$

$$K_\alpha = \frac{A-4}{A} Q$$

For α decay $A > 210$ which means maximum part of released energy is associated with K.E. of α . If Q is negative, the reaction is endoergic. The minimum amount of energy that a bombarding particle must have in order to initiate an endoergic reaction is called Threshold energy E_{th} ,

given by $E_{th} = -Q \left(\frac{m_1}{m_2} + 1 \right)$ where m_1 = mass of the projectile.

E_{th} = minimum kinetic energy of the projectile to initiate the nuclear reaction

m_2 = mass of the target

Ex. How much energy must a bombarding proton possess to cause the reaction ${}_3\text{Li}^7 + {}_1\text{H}^1 \rightarrow {}_4\text{Be}^7 + {}_0n^1$ (Mass of ${}_3\text{Li}^7$ atom is 7.01600, mass of ${}_1\text{H}^1$ atom is 1.0783, mass of ${}_4\text{Be}^7$ atom is 7.01693)

Sol. Since the mass of an atom includes the masses of the atomic electrons, the appropriate number of electron masses must be subtracted from the given values.

Reactants : Total mass = $(7.01600 - 3 m_e) + (1.0783 - 1 m_e) = 8.0943 - 4m_e$

Products : Total mass = $(7.01693 - 4m_e) + 1.0087 = 8.02563 - 4m_e$

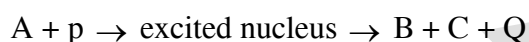
The energy is supplied as kinetic energy of the bombarding proton. The incident proton must have more than this energy because the system must possess some kinetic energy even after the reaction, so that momentum is conserved with momentum conservation taken into account, the minimum kinetic energy that the incident particle must possess can be found with the formula.

where, $Q = - [(8.02563 - 4m_p) - (8.0943 - 4m_p)] \times 931.5 \text{ MeV} = -63.96 \text{ MeV}$

$$E_{th} = - \left(1 + \frac{m}{M}\right) Q = - \left(1 + \frac{1}{7}\right) (-63.96) = 73.1 \text{ MeV}$$

NUCLEAR FISSION

In 1938 by Hahn and Strassmann. By attack of a particle splitting of a heavy nucleus ($A > 230$) into two or more lighter nuclei. In this process certain mass disappears which is obtained in the form of energy (enormous amount)

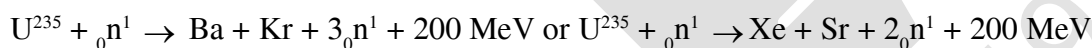


Hahn and Strassmann done the first fission of nucleus of U^{235} .

When U^{235} is bombarded by a neutron it splits into two fragments and 2 or 3 secondary neutrons and releases about 190 MeV ($\approx 200 \text{ MeV}$) energy per fission (or from single nucleus)

Fragments are uncertain but each time energy released is almost same.

Possible reactions are



and many other reactions are possible.

- The average number of secondary neutrons is 2.5.
- Nuclear fission can be explained by using "liquid drop model" also.
- The mass defect Δm is about 0.1% of mass of fissioned nucleus
- About 93% of released energy (Q) is appear in the form of kinetic energies of products and about 7% part in the form of γ - rays.

NUCLEAR CHAIN REACTION :

The equation of fission of U^{235} is $U^{235} + {}_0^1n^1 \rightarrow Ba + Kr + 3{}_0^1n^1 + Q$.

These three secondary neutrons produced in the reaction may cause of fission of three more U^{235} and give 9 neutrons, which in turn, may cause of nine more fission of U^{235} and so on.

Thus a continuous 'Nuclear Chain reaction' would start.

If there is no control on chain reaction then in a short time ($\approx 10^{-6} \text{ sec.}$) a huge amount of energy will be released. (This is the principle of 'Atom bomb'). If chain is controlled then produced energy can be used for peaceful purposes. For example nuclear reactor (Based on fission) are generating electricity.

NATURAL URANIUM :

It is mixture of U^{235} (0.7%) and U^{238} (99.3%).

U^{235} is easily fissionable, by slow neutron (or thermal neutrons) having K.E. of the order of 0.03 eV.

But U^{238} is fissionable with fast neutrons.

Note : Chain reaction in natural uranium can't occur

To improve the quality, percentage of U^{235} is increased to 3%.

The proposed uranium is called 'Enriched Uranium' (97% U^{238} and 3% U^{235})

LOSSES OF SECONDARY NEUTRONS :

Leakage of neutrons from the system : Due to their maximum K.E. some neutrons escape from the system.

Absorption of neutrons by U^{238} : Which is not fissionable by these secondary neutrons.

CRITICAL SIZE (OR MASS) :

In order to sustain chain reaction in a sample of enriched uranium, it is required that the number of lost neutrons should be much smaller than the number of neutrons produced in a fission process. For it the size of uranium block should be equal or greater than a certain size called **critical size**.

REPRODUCTION FACTOR :

$$(K) = \frac{\text{rate of production of neutrons}}{\text{rate of loss of neutrons}}$$

- (i) If size of Uranium used is 'Critical' then $K = 1$ and the chain reaction will be steady or sustained (As in nuclear reaction)
- (ii) If size of Uranium used is 'Super critical' then $K > 1$ and chain reaction will accelerate resulting in a explosion (As in atom bomb)
- (iii) If size of Uranium used is 'Sub Critical' then $K < 1$ and chain reaction will retard and will stop.

NUCLEAR REACTOR ($K = 1$) : Credit $\rightarrow$ To Enricho Fermi

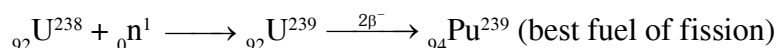
Construction :

- **Nuclear Fuel :** Commonly used are U^{235} , Pu^{239} . Pu^{239} is the **best**. Its critical size is less than critical size of U^{235} .
But Pu^{239} is not naturally available and U^{235} is used in most of the reactors.
- **Moderator :** Its function is to slow down the fast secondary neutrons. Because only slow neutrons can bring the fission of U^{235} . The moderator should be light and it should not absorb the neutrons. Commonly, Heavy water (D_2O , molecular weight 20 gm.) Graphite etc. are used. These are rich of protons. Neutrons collide with the protons and interchange their energy. Thus neutrons get slow down.
- **Control rods :** They have the ability to capture the slow neutrons and can control the chain reaction at any stage. Boron and Cadmium are best absorber of neutrons.
- **Coolant :** A substance which absorb the produced heat and transfers it to water for further use. Generally coolant is water at high pressure

FAST BREADER REACTORS

The atomic reactor in which fresh fissionable fuel (Pu^{239}) is produced along with energy. The amount of produced fuel (Pu^{239}) is more than consumed fuel (U^{235})

- **Fuel :** Natural Uranium.
- **Process:** During fission of U^{235} , energy and secondary neutrons are produced. These secondary neutrons are absorbed by U^{238} and U^{239} is formed. This U^{239} converts into Pu^{239} after two beta decay. This Pu^{239} can be separated, its half life is 2400 years.

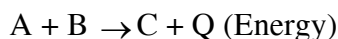


This Pu^{239} can be used in nuclear weapons because of its small critical size than U^{235} .

- **Moderator** : Are not used in these reactors.
- **Coolant** : Liquid sodium

NUCLEAR FUSION :

It is the phenomenon of fusing two or more lighter nuclei to form a single heavy nucleus.



The product (C) is more stable than reactants (A and B) and $m_c < (m_a + m_b)$

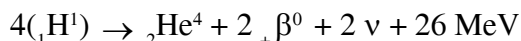
and mass defect $\Delta m = [(m_a + m_b) - m_c] \text{ amu}$

Energy released is $E = \Delta m \times 931 \text{ MeV}$

The total binding energy and binding energy per nucleon C both are more than of A and B.

$$\Delta E = E_c - (E_a + E_b)$$

Fusion of four hydrogen nuclei into helium nucleus :



- Energy released per fission $\gg$ Energy released per fusion
- Energy per nucleon in fission $\left[= \frac{200}{235} \approx 0.85 \text{ MeV} \right] \ll$ energy per nucleon in fusion $\left[= \frac{24}{4} \approx 6 \text{ MeV} \right]$

REQUIRED CONDITION FOR NUCLEAR FUSION

- **High temperature** :
Which provide kinetic energy to nuclei to overcome the repulsive electrostatic force between them.
- **High Pressure (or density)** :
Which ensure frequent collision and increases the probability of fusion. The required temperature and pressure at earth (lab) are not possible. These condition exist in the sun and in many other stars. The source of energy in the sun is nuclear fusion, where hydrogen is in plasma state and there protons fuse to form helium nuclei.

HYDROGEN BOMB

It is based on nuclear fusion and produces more energy than an atom bomb.

Pair production	Pair Annihilation
A γ-photon of energy more than 1.02 MeV, when interact with a nucleus produces pair of electron (e^-) and positron (e^+). The energy equivalent to rest mass of e^- (or e^+) = 0.51 MeV. The energy equivalent to rest mass of pair ($e^- + e^+$) = 1.02 MeV. For pair production Energy of photon $\geq 1.02 \text{ MeV}$. If energy of photon is more than 1.02 MeV, the extra energy $(E - 1.02) \text{ MeV}$ divides approximately in equal amount to each particle as the kinetic energy or $(KE)_{e^- \text{ or } e^+} = \left[\frac{E_{ph} - 1.02}{2} \right] \text{ MeV}$ If $E < 1.02 \text{ MeV}$, pair will not produce.	When electron and positron combines they annihilates to each other and only energy is released in the form of two gamma photons. If the energy of electron and positron are negligible then energy of each γ-photon is 0.51 MeV

Ex. In a nuclear reactor, fission is produced in 1 g for U^{235} (235.0439) in 24 hours by slow neutrons (1.0087 u). Assume that $_{35}\text{Kr}^{92}$ (91.8973 u) and $_{56}\text{Ba}^{141}$ (140.9139 amu) are produced in all reactions and no energy is lost.

(i) Write the complete reaction (ii) Calculate the total energy produced in kilowatt hour.

Given $1u = 931 \text{ MeV}$.

Sol. The nuclear fission reaction is ${}_{92}\text{U}^{235} + {}_0\text{n}^1 \rightarrow {}_{56}\text{Ba}^{141} + {}_{36}\text{Kr}^{92} + 3{}_0\text{n}^1$

$$\text{Mass defect } \Delta m = [(m_u + m_n) - (m_{Ba} + m_{Kr} + 3m_n)] = 256.0526 - 235.8373 = 0.2153 \text{ u}$$

Energy released $Q = 0.2153 \times 931 = 200 \text{ MeV}$. Number of atoms in $1 \text{ g} = \frac{6.02 \times 10^{23}}{235} = 2.56 \times 10^{21}$

Energy released in fission of 1 g of U^{235} is $E = 200 \times 2.56 \times 10^{21} = 5.12 \times 10^{23} \text{ MeV}$
 $= 5.12 \times 10^{23} \times 1.6 \times 10^{-13} = 8.2 \times 10^{10} \text{ J}$

$$= \frac{8.2 \times 10^{10}}{3.6 \times 10^6} \text{ kWh} = 2.28 \times 10^4 \text{ kWh}$$

Ex. It is proposed to use the nuclear fusion reaction : ${}_1\text{H}^2 + {}_1\text{H}^2 \rightarrow {}_2\text{He}^4$ in a nuclear reactor of 200 MW rating. If the energy from above reaction is used with at 25% efficiency in the reactor, how many grams of deuterium will be needed per day. (Mass of ${}_1\text{H}^2$ is 2.0141 u and mass of ${}_2\text{He}^4$ is 4.0026 u)

Sol. Energy released in the nuclear fusion is $Q = \Delta mc^2 = \Delta m(931) \text{ MeV}$ (where Δm is in amu)

$$Q = (2 \times 2.0141 - 4.0026) \times 931 \text{ MeV} = 23.834 \text{ MeV} = 23.834 \times 10^6 \text{ eV}$$

Since efficiency of reactor is 25%

$$\text{So effective energy used} = \frac{25}{100} \times 23.834 \times 10^6 \times 1.6 \times 10^{-19} \text{ J} = 9.534 \times 10^{-13} \text{ J}$$

Since the two deuterium nuclei are involved in a fusion reaction,

therefore, energy released per deuterium is $\frac{9.534 \times 10^{-13}}{2}$.

For 200 MW power per day, number of deuterium nuclei required

$$= \frac{200 \times 10^6 \times 86400}{\frac{9.534 \times 10^{-23}}{2}} = 3.624 \times 10^{25}$$

Since 2g of deuterium constitute 6×10^{23} nuclei, therefore amount of deuterium required is

$$= \frac{2 \times 3.624 \times 10^{25}}{6 \times 10^{23}} = 120.83 \text{ g/day}$$

RADIOACTIVITY

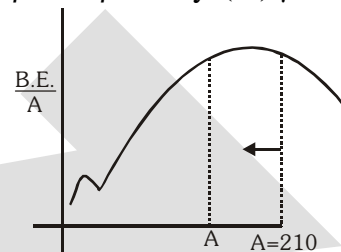
The process of spontaneous disintegration shown by some unstable atomic nuclei is known as natural radioactivity. This property is associated with the emission of certain types of penetrating radiations, called radioactive rays, or Becquerel rays (α , β , γ -rays). The elements or compounds, whose atoms disintegrate and emit radiations are called radioactive elements. Radioactivity is a continuous, irreversible nuclear phenomenon.

Radioactive Decays

Generally, there are three types of radioactive decays (i) α decay (ii) β^- and β^+ decay (iii) γ decay

- **α decay :** In α decay, the unstable nucleus emits an α particle.

By emitting α particle, the nucleus decreases its mass energy number and move towards stability. Nucleus having $A > 210$ shows α decay. By releasing α particle, it can attain higher stability and Q value is positive.



- **β decay :** In beta decay (N/Z) ratio of nucleus is changed. This decay is shown by unstable nuclei. In beta decay, either a neutron is converted into proton or proton is converted into neutron. For better understanding we discuss N/Z graph. There are two type of unstable nuclides

- **A type**

For A type nuclides $(N/Z)_A > (N/Z)_{\text{stable}}$

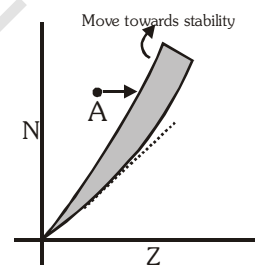
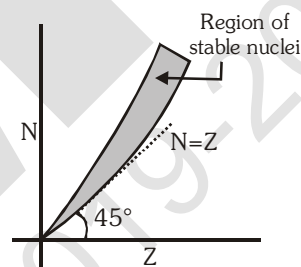
To achieve stability, it increases Z by conversion of neutron into proton



This decay is called β^- decay.

Kinetic energy available for e^{-1} and $\bar{\nu}$ is, $Q = K_{\beta} + K_{\bar{\nu}}$

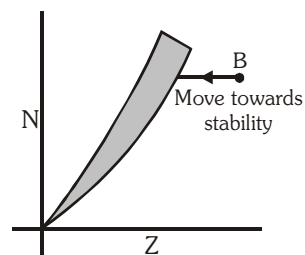
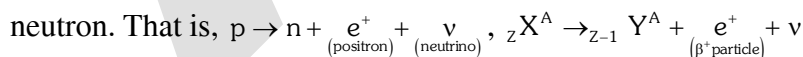
K.E. of β^- satisfies the condition $0 < K_{\beta} < Q$



- **B type**

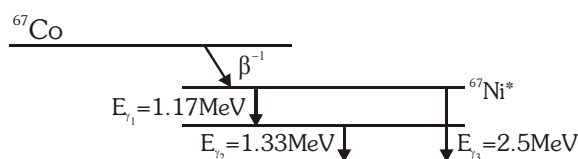
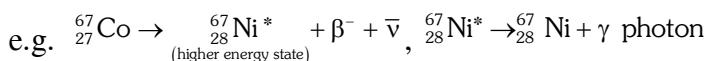
For B type nuclides $(N/Z)_B < (N/Z)_{\text{stable}}$

To achieve stability it decreases Z by the conversion of a proton into



- **γ decay :** When an α or β decay takes place, the daughter nucleus is usually in higher energy state, such a nucleus comes to ground state by emitting a photon or photons.

Order of energy of γ photon is 100 KeV



Properties of α , β and γ rays

Features	α -particles	β -particles	γ -rays
Identity	Helium nucleus or doubly ionised helium atom (${}_2\text{He}^4$)	Fast moving electrons (${}_{-1}\beta^0$ or β^-)	Electromagnetic wave (photons)
Charge	Twice of proton ($+2e$) $\approx 4m_p$	Electronic ($-e$)	Neutral
Mass	m_p —mass of proton	(rest mass of β) = (rest mass of electron)	rest mass = 0
Speed	1.4×10^7 m/s. to 2.2×10^7 m/s. (Only certain value between this range). Their speed depends on nature of the nucleus. So that it is a characteristic speed.	1% of c to 99% of c (All possible values between this range) β -particles come out with different speeds from the same type of nucleus. So that it can not be a characteristic speed.	Only $c = 3 \times 10^8$ m/s γ -photons come out with same speed from all types of nucleus. So, can not be a characteristic speed.
K.E.	$\approx$ MeV	$\approx$ MeV	$\approx$ MeV
Energy spectrum	Line and discrete (or linear)	Continuous (or linear)	Line and discrete
Ionization power ($\alpha > \beta > \gamma$)	10,000 times of γ -rays	100 times of γ -rays (or $\frac{1}{100}$ times of α)	1 (or $\frac{1}{100}$ times of β)
Penetration power ($\gamma > \beta > \alpha$)	$\frac{1}{10000}$ times of γ -rays	$\frac{1}{100}$ times of γ -rays (100 times of α)	1 (100 times of β)
Effect of electric or magnetic field	Deflection	Deflection (More than α)	No deflection
Explanation of emission	By Tunnel effect (or quantum mechanics)	By weak nuclear interactions	With the help of energy levels in nucleus

Laws of Radioactive Decay

- The radioactive decay is a spontaneous process with the emission of α , β and γ rays. It is not influenced by external conditions such as temperature, pressure, electric and magnetic fields.
- The rate of disintegration is directly proportional to the number of radioactive atoms present at that time i.e., rate of decay $\propto$ number of nuclei.

$$\text{Rate of decay} = \lambda (\text{number of nuclei}) \text{ i.e. } \frac{dN}{dt} = -\lambda N$$

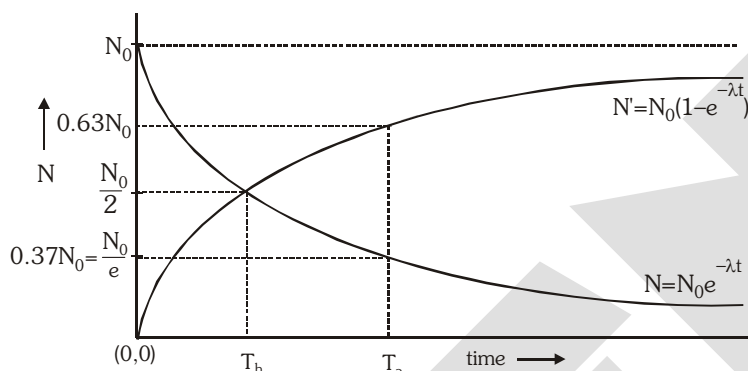
where λ is called the decay constant. This equation may be expressed in the form $\frac{dN}{N} = -\lambda dt$.

$$\int_{N_0}^N \frac{dN}{N} = -\lambda \int_0^t dt \Rightarrow \ln\left(\frac{N}{N_0}\right) = -\lambda t$$

where N_0 is the number of parent nuclei at $t=0$. The number that survives at time t is therefore

$$N = N_0 e^{-\lambda t} \text{ and } t = \frac{2.303}{\lambda} \log_{10}\left(\frac{N_0}{N_t}\right) \text{ this function is plotted in figure.}$$

Graph : Time versus N (or N')



- **Half life (T_h) :** It is the time during which number of active nuclei reduce to half of initial value.

If at $t = 0$ no. of active nuclei N_0 then at $t = T_h$ number of active nuclei will be $\frac{N_0}{2}$

From decay equation $N = N_0 e^{-\lambda t}$

$$\frac{N_0}{2} = N_0 e^{-\lambda T_h} \Rightarrow T_h = \frac{\ln(2)}{\lambda} = \frac{0.693}{\lambda} \approx \frac{0.7}{\lambda}$$

- **Mean or Average Life (T_a) :** It is the average of age of all active nuclei i.e.

$$T_a = \frac{\text{sum of times of existence of all nuclei in a sample}}{\text{initial number of active nuclei in that sample}} = \frac{1}{\lambda}$$

- At $t = 0$, number of active nuclei = N_0 then number of active nuclei at

$$t = T_a \text{ is } N = N_0 e^{-\lambda T_a} = N_0 e^{-1} = \frac{N_0}{e} = 0.37 N_0 = 37\% \text{ of } N_0$$

- Number nuclei which have been disintegrated within duration T_a is

$$N' = N_0 - N = N_0 - 0.37 N_0 = 0.63 N_0 = 63\% \text{ of } N_0$$

$$T_a = \frac{1}{\lambda} = \frac{T_h}{\ln(2)} = \frac{T_h}{0.693} = 1.44 T_h$$

- Within duration $T_h \Rightarrow 50\%$ of N_0 decayed and 50% of N_0 remains active
- Within duration $T_a \Rightarrow 63\%$ of N_0 decayed and 37% of N_0 remains active

ACTIVITY OF A SAMPLE (A OR R) (OR DECAY RATE)

It is the rate of decay of a radioactive sample $R = -\frac{dN}{dt} = N\lambda$ or $R = R_0 e^{-\lambda t}$

- Activity of a sample at any instant depends upon number of active nuclei at that instant.
 $R \propto N$ (or active mass), $R \propto m$
- R also decreases exponentially w.r.t. time same as the number of active nuclei decreases.

- R is not a constant with N, m and time while λ , T_h and T_a are constant
- At $t = 0$, $R = R_0$ then at $t = T_h \Rightarrow R = \frac{R_0}{2}$ and at $t = T_a \Rightarrow R = \frac{R_0}{e}$ or $0.37 R_0$
- Similarly active mass of radioactive sample decreases exponentially. $m = m_0 e^{-\lambda t}$
- Activity of m gm active sample (molecular weight M_w) is $R = \lambda N = \frac{0.693}{T_h} \left[\frac{N_{AV}}{M_w} \right] m$

here N_{AV} = Avogadro number = 6.023×10^{23}

SI UNIT of R : 1 becquerel (1 Bq) = 1 decay/sec

Other Unit is curie : 1 Ci = 3.70×10^{10} decays/sec

1 Rutherford : (1 Rd) = 10^6 decays/s

Specific activity : Activity of 1 gm sample of radioactive substance. Its unit is Ci/gm
e.g. specific activity of radium (226) is 1 Ci/gm.

Ex. The half-life of cobalt-60 is 5.25 yrs. After how long does its activity reduce to about one eighth of its original value?

Sol The activity is proportional to the number of undecayed atoms: In each half-life, the remaining sample decays to half of its initial value. Since $\left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) = \frac{1}{8}$, therefore, three half-lives or 15.75 years are required for the sample to decay to 1/8th its original strength.

Ex. A count rate meter is used to measure the activity of a given sample. At one instant the meter shows 4750 counts per minute. Five minutes later it shows 2700 counts per minute.
(i) Find the decay constant. (ii) Also, find the half-life of the sample.

Sol. Initial activity $A_i = \left. \frac{-dN}{dt} \right|_{t=0} = \lambda N_0 = 4750$... (i) Final activity $A_f = \left. \frac{-dN}{dt} \right|_{t=5} = \lambda N = 2700$... (ii)

Dividing (i) by (ii), we get $\frac{4750}{2700} = \frac{N_0}{N_t}$

The decay constant is given by $\lambda = \frac{2.303}{t} \log \frac{N_0}{N_t} = \frac{2.303}{5} \log \frac{4750}{2700} = 0.113 \text{ min}^{-1}$

Half-life of the sample is $T = \frac{0.693}{\lambda} = \frac{0.693}{0.113} = 6.14 \text{ min}$

• Parallel radioactive disintegration

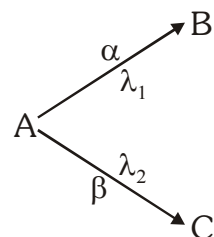
Let initial number of nuclei of A is N_0 then at any time number of nuclei of

A, B & C are given by $N_0 = N_A + N_B + N_C \Rightarrow \frac{dN_A}{dt} = -\frac{d}{dt}(N_B + N_C)$

A disintegrates into B and C by emitting α , β particle.

Now, $\frac{dN_B}{dt} = -\lambda_1 N_A$ and $\frac{dN_C}{dt} = -\lambda_2 N_A \Rightarrow \frac{d}{dt}(N_B + N_C) = -(\lambda_1 + \lambda_2) N_A$

$\Rightarrow \frac{dN_A}{dt} = -(\lambda_1 + \lambda_2) N_A \Rightarrow \lambda_{\text{eff}} = \lambda_1 + \lambda_2 \Rightarrow t_{\text{eff}} = \frac{t_1 t_2}{t_1 + t_2}$



Ex. The mean lives of a radioactive substances are 1620 and 405 years for α -emission and β -emission respectively. Find out the time during which three fourth of a sample will decay if it is decaying both by α -emission and β -emission simultaneously.

Sol. When a substance decays by α and β emission simultaneously, the average rate of disintegration λ_{av} is given by

$\lambda_{av} = \lambda_{\alpha} + \lambda_{\beta}$ when λ_{α} = disintegration constant for α -emission only λ_{β} = disintegration constant for β -emission only

Mean life is given by $T_m = \frac{1}{\lambda}$, $\lambda_{av} = \lambda_{\alpha} + \lambda_{\beta} \Rightarrow \frac{1}{T_m} = \frac{1}{T_{\alpha}} + \frac{1}{T_{\beta}} = \frac{1}{1620} + \frac{1}{405} = \frac{1}{324}$

$\lambda_{av} \times t = 2.303 \log \frac{N_0}{N_t}$, $\frac{1}{324} t = 2.303 \log \frac{100}{25} \Rightarrow t = 2.303 \times 324 \log 4 = 449$ years.

Ex. A radioactive decay is given by $A \xrightarrow{t_{1/2}=8 \text{ yrs}} B$

Only A is present at $t=0$. Find the time at which if we are able to pick one atom out of the sample, then probability of getting B is 15 times of getting A.

Sol.

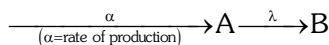
	A	$\longrightarrow$	B
at $t=0$	N_0		0
at $t=t$	N		$N_0 - N$

Probability of getting A, $P_A = \frac{N}{N_0}$

Probability of getting B, $P_B = \frac{N_0 - N}{N_0} \Rightarrow P_B = 15 P_A \Rightarrow \frac{N_0 - N}{N_0} = 15 \frac{N}{N_0} \Rightarrow N_0 = 16N \Rightarrow N = \frac{N_0}{16}$

Remaining nuclei are $\frac{1}{16}$ th of initial nuclei, hence required time $t = 4$ half lives = 32 years

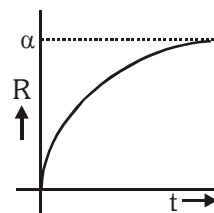
Radioactive Disintegration with Successive Production



$$\frac{dN_A}{dt} = \alpha - \lambda N_A \dots (i)$$

when N_A in maximum $\frac{dN_A}{dt} = 0 = \alpha - \lambda N_A = 0$, $N_A \text{ max} = \frac{\alpha}{\lambda} = \frac{\text{rate of production}}{\lambda}$

By equation (i) $\int_0^t \frac{dN_A}{\alpha - \lambda N_A} = \int_0^t dt$, Number of nuclei is $N_A = \frac{\alpha}{\lambda} (1 - e^{-\lambda t})$



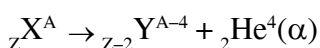
Ex. $10^{21} \text{ per sec} \rightarrow A \xrightarrow{\lambda = \frac{1}{30}} B$

A shows radioactive disintegration and it is continuously produced at the rate of 10^{21} per sec. Find maximum number of nuclei of A.

Sol. At maximum, $r_{\text{production}} = r_{\text{decay}} \Rightarrow 10^{21} = \frac{1}{30} N \Rightarrow N = 30 \times 10^{21}$

Soddy and Fajan's Group Displacement Laws :

(i) **α -decay :** The emission of one α -particle reduces the mass number by 4 units and atomic number by 2 units. If parent and daughter nuclei are represented by symbols X and Y respectively then,

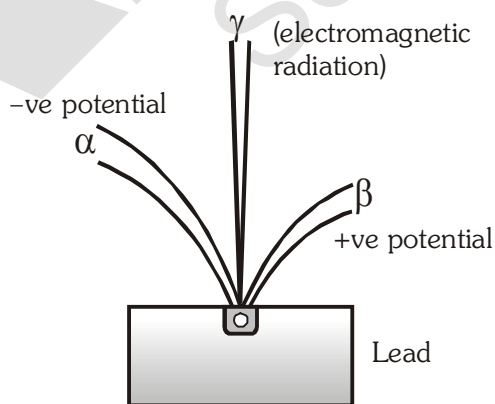


(ii) **β -decay :** Beta particles are said to be fast moving electrons coming from the nucleus of a radioactive substance. Does it mean that a nucleus contains electrons? No, it is an established fact that nucleus does not contain any electrons. When a nucleus emits a beta particle, one of its neutrons breaks into a proton, an electron (i.e., β -particle) and an antineutrino $n \rightarrow p + e + \bar{\nu}$

where $n = \text{neutron}$ $p = \text{proton}$ $e = \beta\text{-particle}$

Thus emission of a beta particle is caused by the decay of a neutron into a proton. The daughter nucleus thus has an atomic number greater than one (due to one new proton in the nucleus) but same mass number as that of parent nucleus. Therefore, representing the parent and daughter nucleus by symbols X and Y respectively, we have ${}_Z X^A \rightarrow {}_{Z+1} Y^A + \beta + \bar{\nu}$

(iii) **γ -decay :** When parent atoms emit gamma rays, no charge is involved as these are neutral rays. Thus there is no effect on the atomic number and mass number of the parent nucleus. However the emission of γ -rays represents energy. Hence the emission of these rays changes the nucleus from an excited (high energy) state to a less excited (lower energy) state.



EXERCISE (S-1)

Nuclear Physics :

1. Suppose that the Sun consists entirely of hydrogen atom and releases the energy by the nuclear reaction, $4 {}_1^1\text{H} \longrightarrow {}_2^4\text{He}$ with 26 MeV of energy released. If the total output power of the Sun is assumed to remain constant at 3.9×10^{26} W, find the time it will take to burn all the hydrogen. Take the mass of the Sun as 1.7×10^{30} kg.
2. When two deuterons (${}_1^2\text{H}$) fuse to form a helium nucleus ${}_2^4\text{He}$, 23.6 MeV energy is released. Find the binding energy of helium if it is 1.1 MeV for each nucleon of deuterium.
3. Highly energetic electrons are bombarded on a target of an element containing 30 neutrons. The ratio of radii of nucleus to that of helium nucleus is $(14)^{1/3}$. Find [JEE 2005]
(a) atomic number of the nucleus
(b) the frequency of K_α line of the X-ray produced. ($R = 1.1 \times 10^7 \text{ m}^{-1}$ and $c = 3 \times 10^8 \text{ m/s}$)

Radioactivity :

4. The kinetic energy of an α - particle which flies out of the nucleus of a Ra^{226} atom in radioactive disintegration is 4.78 MeV. Find the total energy evolved during the escape of the α - particle.
5. U^{238} and U^{235} occur in nature in an atomic ratio 140 : 1. Assuming that at the time of earth's formation the two isotopes were present in equal amounts. Calculate the age of the earth. (Half life of $\text{U}^{238} = 4.5 \times 10^9$ yrs & that of $\text{U}^{235} = 7.13 \times 10^8$ yrs)
6. At $t = 0$, a sample is placed in a reactor. An unstable nuclide is produced at a constant rate R in the sample by neutron absorption. This nuclide β^- decays with half life τ . Find the time required to produce 80% of the equilibrium quantity of this unstable nuclide.

EXERCISE (S-2)

1. An experiment is done to determine the half-life of radioactive substance that emits one β -particle for each decay process. Measurement show that 8.4 β are emitted each second by 2.5 mg of the substance. The atomic weight of the substance is 230. Find the half life of the substance.
2. A wooden piece of great antiquity weighs 50 gm and shows C^{14} activity of 320 disintegrations per minute. Estimate the length of the time which has elapsed since this wood was part of living tree, assuming that living plants show a C^{14} activity of 12 disintegrations per minute per gm. The half life of C^{14} is 5730 yrs.
3. A body of mass m_0 is placed on a smooth horizontal surface. The mass of the body is decreasing exponentially with disintegration constant λ . Assuming that the mass is ejected backward with a relative velocity u . Initially the body was at rest. Find the velocity of body after time t .
4. A radionuclide with disintegration constant λ is produced in a reactor at a constant rate α nuclei per sec. During each decay energy E_0 is released. 20% of this energy is utilised in increasing the temperature of water. Find the increase in temperature of m mass of water in time t . Specific heat of water is S . Assume that there is no loss of energy through water surface.
5. The element Curium $^{248}_{96}\text{Cm}$ has a mean life of 10^{13} seconds. Its primary decay modes are spontaneous fission and α decay, the former with a probability of 8% and the latter with a probability of 92%. Each fission releases 200 MeV of energy. The masses involved in α decay are as follows
 $^{248}_{96}\text{Cm} = 248.072220\text{u}$, $^{244}_{94}\text{Pu} = 244.064100\text{u}$ & $^4_2\text{He} = 4.002603\text{u}$.

Calculate the power output from a sample of 10^{20} Cm atoms. ($1\text{ u} = 931\text{ MeV}/c^2$)

6. A small quantity of solution containing ^{24}Na radionuclide (half life 15 hours) of activity 1.0 microcurie is injected into the blood of a person. A sample of the blood of volume 1 cm^3 taken after 5 hours shows an activity of 296 disintegration per minute. Determine the total volume of blood in the body of the person. Assume that the radioactive solution mixes uniformly in the blood of the person.
 (1 Curie = 3.7×10^{10} disintegrations per second)
7. The positron is a fundamental particle with the same mass as that of the electron and with a charge equal to that of an electron but of opposite sign. When a positron and an electron collide, they may annihilate each other. The energy corresponding to their mass appears in two photons of equal energy. Find the wavelength of the radiation emitted.
 [Take : mass of electron = $(0.5/C^2)\text{MeV}$ and $hc = 1.2 \times 10^{-12}\text{ MeV.m}$ where h is the Plank's constant and C is the velocity of light in air]

8. A π^+ meson of negligible initial velocity decays to a μ^+ (muon) and a neutrino. With what kinetic energy (in eV) does the muon move? (The rest mass of neutrino can be considered zero. The rest mass of the π^+ meson is 150 MeV and the rest mass of the muon is 100 MeV.) Take neutrino to behave like a photon. Take $\sqrt{2} = 1.41$.
9. A radioactive decay counter is switched on at $t = 0$. A β - active sample is present near the counter. The counter registers the number of β -particles emitted by the sample. The counter registers 1×10^5 β -particles at $t = 36$ s and 1.11×10^5 β -particles at $t = 108$ s. Find $T_{1/2}$ of this sample
10. A radioactive sample emits n β -particles in 2 sec. In next 2 sec it emits $0.75 n$ β -particles, what is the mean life of the sample? [JEE 2003]
11. The age of a rock containing lead and uranium is equal to 1.5×10^9 yrs. The uranium is decaying into lead with half life equal to 4.5×10^9 yrs. Find the ratio of lead to uranium present in the rock, assuming initially no lead was present in the rock. (Given $2^{1/3} = 1.259$). [JEE 2004]

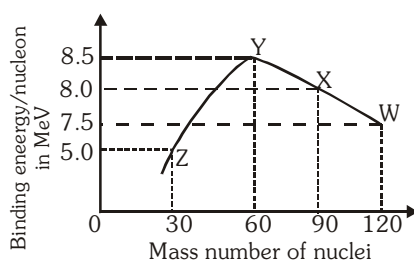
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Nuclear Physics :

- In Rutherford's famous gold foil scattering experiment, he found that most alpha particles would pass through the foil undeflected. Which one of the following nuclear properties can be inferred from this observation?
(A) The nucleus must have a positive charge
(B) Most of the mass of an atom is in the nucleus
(C) The nucleus contains both protons neutrons
(D) The diameter of the nucleus is small compared to the diameter of the atom
(E) None of these
- Let u denote one atomic mass unit. One atom of an element of mass number A has mass exactly equal to Au
(A) for any value of A
(B) only for $A = 1$
(C) only for $A = 12$
(D) for any value of A provided the atom is stable
- If radius of the ${}^{27}_{13}\text{Al}$ nucleus is estimated to be 3.6 fermi, then the radius of ${}^{125}_{52}\text{Te}$ nucleus be nearly-
[AIEEE - 2005]
(A) 6 fermi
(B) 8 fermi
(C) 4 fermi
(D) 5 fermi
- The surface area of a nucleus varies with mass number A as
(A) $A^{2/3}$
(B) $A^{1/3}$
(C) A
(D) None
- A nucleus disintegrates into two nuclear parts which have their velocities in the ratio 2 : 1. The ratio of their nuclear sizes will be-
[AIEEE - 2004]
(A) $2^{1/3} : 1$
(B) $1 : 3^{1/2}$
(C) $3^{1/2} : 1$
(D) $1 : 2^{1/3}$
- The binding energy per nucleon for C^{12} is 7.68 MeV and that for C^{13} is 7.5 MeV. The energy required to remove a neutron from C^{13} is
(A) 5.34 MeV
(B) 5.5 MeV
(C) 9.5 MeV
(D) 9.34 MeV
- The following nuclear reaction is an example of ${}^{12}_6\text{C} + {}^4_2\text{H} \rightarrow {}^{16}_8\text{O} + \text{energy}$
(A) fission
(B) fusion
(C) alpha decay
(D) beta decay
- Fast neutrons may most easily be slowed down by which one of the following methods?
(A) passing them through a substance rich in hydrogen
(B) allowing them to collide elastically with heavy nuclei
(C) using lead shielding
(D) passing them through an increasing potential gradient space

9. A nuclear transformation is denoted by $X(n, \alpha) \rightarrow {}^7_3\text{Li}$. Which of the following is the nucleus of element X ? [AIEEE - 2005]
- (A) ${}^{12}_6\text{C}$ (B) ${}^{10}_5\text{B}$ (C) ${}^9_5\text{B}$ (D) ${}^{11}_4\text{Be}$
10. A certain radioactive nuclide of mass number m_x disintegrates, with the emission of an electron and γ radiation only, to give second nuclide of mass number m_y . Which one of the following equation correctly relates m_x and m_y ?
- (A) $m_y = m_x + 1$ (B) $m_y = m_x - 2$ (C) $m_y = m_x - 1$ (D) $m_y = m_x$
11. A nucleus with $Z = 92$ emits the following in a sequence : $\alpha, \alpha, \beta^-, \beta^-, \alpha, \alpha, \alpha, \alpha, \beta^-, \beta^-, \alpha, \beta^+, \beta^+, \alpha$. The Z of the resulting nucleus is- [AIEEE - 2003]
- (A) 76 (B) 78 (C) 82 (D) 74
12. In an α -decay the Kinetic energy of α particle is 48 MeV and Q -value of the reaction is 50 MeV. The mass number of the mother nucleus is: (Assume that daughter nucleus is in ground state)
- (A) 96 (B) 100 (C) 104 (D) none of these
13. When U^{238} nucleus originally at rest, decays by emitting an alpha particle having a speed u , the recoil speed of the residual nucleus is- [AIEEE - 2003]
- (A) $\frac{4u}{238}$ (B) $-\frac{4u}{234}$ (C) $\frac{4u}{234}$ (D) $-\frac{4u}{238}$
14. If a star converts all of its Helium into oxygen nucleus, find the amount of energy released per nucleus of oxygen. $\text{O} = 15.9994 \text{ amu}$ and $\text{He} = 4.0026 \text{ amu}$ [JEE' 2005 (Scr)]
- (A) 7.26 MeV (B) 7 MeV (C) 10.24 MeV (D) 5.12 MeV
15. The nucleus of element X ($A = 220$) undergoes α -decay. If Q -value of the reaction is 5.5 MeV, then the kinetic energy of α -particle is : [JEE 2003 (Scr)]
- (A) 5.4 MeV (B) 10.8 MeV (C) 2.7 MeV (D) None
16. In the uranium radioactive series the initial nucleus is ${}_{92}\text{U}^{238}$, and the final nucleus is ${}_{82}\text{Pb}^{206}$. When the uranium nucleus decays to lead, the number of α - particles emitted is.. and the number of β -particles emitted...
- (A) 6, 8 (B) 8, 6 (C) 16, 6 (D) 32, 12
17. Binding energy per nucleon versus mass number curve for nuclei is shown in figure. W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is :

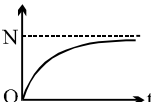
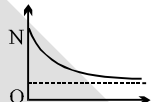
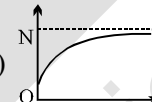
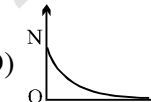


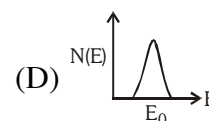
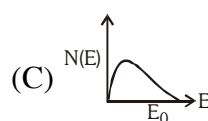
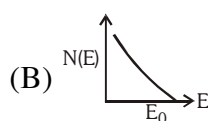
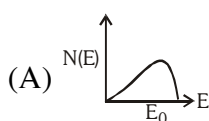
- (A) $Y \rightarrow 2Z$ (B) $W \rightarrow X + Z$ (C) $W \rightarrow 2Y$ (D) $X \rightarrow Y + Z$

18. An alpha nucleus of energy $\frac{1}{2}mv^2$ bombards a heavy nuclear target of charge Ze . Then the distance of closest approach for the alpha nucleus will be proportional to- [AIEEE - 2006]
 (A) v^2 (B) $1/m$ (C) $1/v^4$ (D) $1/Ze$
19. If the binding energy per nucleon in ${}^7_3\text{Li}$ and ${}^4_2\text{He}$ nuclei are 5.60 MeV and 7.06 MeV respectively, then in the reaction : $p + {}^7_3\text{Li} \rightarrow 2{}^4_2\text{He}$ energy of proton must be- [AIEEE - 2006]
 (A) 28.24 MeV (B) 17.28 MeV (C) 1.46 MeV (D) 39.2 MeV
20. If M_0 is the mass of an oxygen isotope ${}^{17}_8\text{O}$, M_p and M_n are the masses of a proton and a neutron, respectively, the nuclear binding energy of the isotope is- [AIEEE - 2007]
 (A) $(M_0 - 8M_p)c^2$ (B) $(M_0 - 8M_p - 9M_n)c^2$
 (C) M_0c^2 (D) $(M_0 - 17M_n)c^2$
21. In the options given below, let E denote the rest mass energy of a nucleus and n a neutron. The correct option is :- [JEE 2007]
 (A) $E({}^{236}_{92}\text{U}) > E({}^{137}_{53}\text{I}) + E({}^{97}_{39}\text{Y}) + 2E(n)$ (B) $E({}^{236}_{92}\text{U}) < E({}^{137}_{53}\text{I}) + E({}^{97}_{39}\text{Y}) + 2E(n)$
 (C) $E({}^{236}_{92}\text{U}) < E({}^{140}_{56}\text{Ba}) + E({}^{94}_{36}\text{Kr}) + 2E(n)$ (D) $E({}^{236}_{92}\text{U}) = E({}^{140}_{56}\text{Ba}) + E({}^{94}_{36}\text{Kr}) + 2E(n)$

Radioactivity :

22. The half-life of ${}^{131}\text{I}$ is 8 days. Given a sample of ${}^{131}\text{I}$ at time $t = 0$, we can assert that :
 (A) no nucleus will decay before $t = 4$ days (B) no nucleus will decay before $t = 8$ days
 (C) all nuclei will decay before $t = 16$ days (D) a given nucleus may decay at any time after $t = 0$
23. Activity of a radioactive substance is R_1 at time t_1 and R_2 at time t_2 ($t_2 > t_1$). Then the ratio $\frac{R_2}{R_1}$ is:
 (A) $\frac{t_2}{t_1}$ (B) $e^{-\lambda(t_1+t_2)}$ (C) $e^{\left(\frac{t_1-t_2}{\lambda}\right)}$ (D) $e^{\lambda(t_1-t_2)}$
24. A particular nucleus in a large population of identical radioactive nuclei did survive 5 half lives of that isotope. Then the probability that this surviving nucleus will survive the next half life :
 (A) $\frac{1}{32}$ (B) $\frac{1}{5}$ (C) $\frac{1}{2}$ (D) $\frac{1}{10}$
25. A certain radio active substance has a half life of 5 years. Thus for a nucleus in a sample of the element, the probability of decay in ten years is
 (A) 50% (B) 75% (C) 100% (D) 60%
26. The activity of a sample reduces from A_0 to $A_0/\sqrt{3}$ in one hour. The activity after 3 hours more will be :-
 (A) $\frac{A_0}{3\sqrt{3}}$ (B) $\frac{A_0}{9}$ (C) $\frac{A_0}{9\sqrt{3}}$ (D) $\frac{A_0}{27}$

27. Half life of radium is 1620 years. How many radium nuclei decay in 5 hours in 5 gm radium? (Atomic weight of radium = 223)
 (A) 9.1×10^{12} (B) 3.23×10^{15} (C) 1.72×10^{20} (D) 3.3×10^{17}
28. The activity of a sample of radioactive material is A_1 at time t_1 and A_2 at time t_2 ($t_2 > t_1$). Its mean life is T .
 (A) $A_1 t_1 = A_2 t_2$ (B) $\frac{A_1 - A_2}{t_2 - t_1} = \text{constant}$
 (C) $A_2 = A_1 e^{(t_1 - t_2)/T}$ (D) $A_2 = A_1 e^{(t_1/Tt_2)}$
29. The decay constant of the end product of a radioactive series is
 (A) zero (B) infinite
 (C) finite (non zero) (D) depends on the end product.
30. A radioactive substance is dissolved in a liquid and the solution is heated. The activity of the solution
 (A) is smaller than that of element
 (B) is greater than that of element
 (C) is equal to that of element
 (D) will be smaller or greater depending upon whether the solution is weak or concentrated.
31. In a certain nuclear reactor, a radioactive nucleus is being produced at a constant rate = 1000 /s. The mean life of the radionuclide is 40 minutes. At steady state, the number of radionuclide will be
 (A) 4×10^4 (B) 24×10^4 (C) 24×10^5 (D) 24×10^6
32. In the above question, if there were 20×10^5 radionuclide at $t = 0$, then the graph of N v/s t is
 (A)  (B)  (C)  (D) 
33. Which of the following cannot be emitted by radioactive substances during their decay ?
 [AIEEE - 2003]
 (A) Protons (B) Neutrinos (C) Helium nuclei (D) Electrons
34. A 280 days old radioactive substance shows an activity of 6000 dps, 140 days later it's activity becomes 3000dps. What was its initial activity.
 [JEE 2004 (Scr)]
 (A) 20000 dps (B) 24000 dps (C) 12000 dps (D) 6000 dps
35. When ${}_3\text{Li}^7$ nuclei are bombarded by protons and the resultant nuclei are ${}_4\text{Be}^8$, the emitted particles will be-
 [AIEEE - 2006]
 (A) alpha particles (B) beta particles (C) gamma photons (D) neutrons
36. The energy spectrum of β -particles [number $N(E)$ as a function of β -energy E] emitted from a radioactive source is-
 [AIEEE - 2006]



37. In gamma ray emission from a nucleus [AIEEE-2007]
 (A) both the neutron number and the proton number change
 (B) there is no change in the proton number and the neutron number
 (C) only the neutron number changes
 (D) only the proton number changes
38. The half-life period of a radioactive element X is same as the mean life time of another radioactive element Y. Initially they have the same number of atoms. Then- [AIEEE - 2007]
 (A) X will decay faster than Y (B) Y will decay faster than X
 (C) Y and X have same decay rate initially (D) X and Y decay at same rate always
39. Given a sample of Radium-226 having half-life of 4 days. Find the probability, a nucleus disintegrates within 2 half lives. [JEE 2006]
 (A) 1 (B) 1/2 (C) 3/4 (D) 1/4

MATRIX MATCH TYPE QUESTION

40. Match the following Columns :- [JEE 2006]

Column-I

- (A) Nuclear fusion
 (B) Nuclear fission
 (C) β -decay
 (D) Exothermic nuclear reaction

Column-II

- (P) Converts some matter into energy
 (Q) Generally occurs for nuclei with low atomic number
 (R) Generally occurs for nuclei with higher atomic number
 (S) Essentially proceeds by weak nuclear forces

41. Some laws/processes are given in **Column I**. Match these with the physical phenomena given in **Column II** and indicate your answer by darkening appropriate bubbles in the 4×4 matrix given in the ORS. [JEE 2007]

Column-I

- (A) Transition between two atomic energy levels
 (B) Electron emission from a material
 (C) Mosley's law
 (D) Change of photon energy into kinetic energy of electrons

Column-II

- (P) Characteristic X-rays
 (Q) Photoelectric effect
 (R) Hydrogen spectrum
 (S) β -decay

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

- The binding energies of nuclei X and Y are E_1 and E_2 respectively. Two atoms of X fuse to give one atom of Y and an energy Q is released. Then:
 (A) $Q = 2E_1 - E_2$ (B) $Q = E_2 - 2E_1$ (C) $Q = 2E_1 + E_2$ (D) $Q = 2E_2 + E_1$
- If each fission in a U^{235} nucleus releases 200 MeV, how many fissions must occur per second to produce a power of 1 KW
 (A) 1.325×10^{13} (B) 3.125×10^{13} (C) 1.235×10^{13} (D) 2.135×10^{13}
- The binding energies of the atom of elements A & B are E_a & E_b respectively. Three atoms of the element B fuse to give one atom of element A. This fusion process is accompanied by release of energy e. Then E_a , E_b are related to each other as
 (A) $E_a + e = 3E_b$ (B) $E_a = 3E_b$ (C) $E_a - e = 3E_b$ (D) $E_a + 3E_b + e = 0$
- The rest mass of the deuteron, ${}^2_1\text{H}$, is equivalent to an energy of 1876 MeV, the rest mass of a proton is equivalent to 939 MeV and that of a neutron to 940 MeV. A deuteron may disintegrate to a proton and a neutron if it :
 (A) emits a γ -ray photon of energy 2 MeV (B) captures a γ -ray photon of energy 2 MeV
 (C) emits a γ -ray photon of energy 3 MeV (D) captures a γ -ray photon of energy 3 MeV
- The number of α and β^- emitted during the radioactive decay chain starting from ${}^{226}_{88}\text{Ra}$ and ending at ${}^{206}_{82}\text{Pb}$ is
 (A) 3α & $6\beta^-$ (B) 4α & $5\beta^-$ (C) 5α & $4\beta^-$ (D) 6α & $6\beta^-$
- In a radioactive element the fraction of initial amount remaining after its mean life time is
 (A) $1 - \frac{1}{e}$ (B) $\frac{1}{e^2}$ (C) $\frac{1}{e}$ (D) $1 - \frac{1}{e^2}$
- Two radioactive material A_1 and A_2 have decay constants of $10\lambda_0$ and λ_0 . If initially they have same number of nuclei, the ratio of number of their undecayed nuclei will be $(1/e)$ after a time
 (A) $\frac{1}{\lambda_0}$ (B) $\frac{1}{9\lambda_0}$ (C) $\frac{1}{10\lambda_0}$ (D) 1
- 90% of a radioactive sample is left undecayed after time t has elapsed. What percentage of the initial sample will decay in a total time 2t:
 (A) 20% (B) 19% (C) 40% (D) 38%

9. A radioactive material of half-life T was produced in a nuclear reactor at different instants, the quantity produced second time was twice of that produced first time. If now their present activities are A_1 and A_2 respectively then their age difference equals:

(A) $\frac{T}{\ln 2} \left| \ln \frac{A_1}{A_2} \right|$ (B) $T \left| \ln \frac{A_1}{A_2} \right|$ (C) $\frac{T}{\ln 2} \left| \ln \frac{A_2}{2A_1} \right|$ (D) $T \left| \ln \frac{A_2}{2A_1} \right|$

10. The half-life of substance X is 45 years, and it decomposes to substance Y. A sample from a meteorite was taken which contained 2% of X and 14% of Y by quantity of substance. If substance Y is not normally found on a meteorite, what is the approximate age of the meteorite?

(A) 270 years (B) 135 years (C) 90 years (D) 45 years

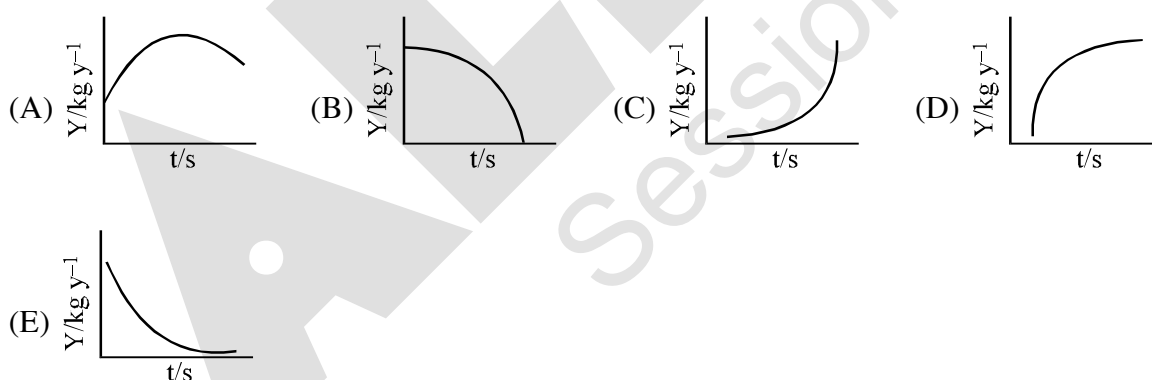
11. At time $t = 0$, N_1 nuclei of decay constant λ_1 & N_2 nuclei of decay constant λ_2 are mixed. The decay rate of the mixture is :

(A) $N_1 N_2 e^{-(\lambda_1 + \lambda_2)t}$ (B) $\left(\frac{N_1}{N_2} \right) e^{-(\lambda_1 - \lambda_2)t}$
(C) $(N_1 \lambda_1 e^{-\lambda_1 t} + N_2 \lambda_2 e^{-\lambda_2 t})$ (D) $N_1 \lambda_1 N_2 \lambda_2 e^{-(\lambda_1 + \lambda_2)t}$

12. A radioactive nuclide can decay simultaneously by two different processes which have decay constants λ_1 and λ_2 . The effective decay constant of the nuclide is λ , then :

(A) $\lambda = \lambda_1 + \lambda_2$ (B) $\lambda = 1/2(\lambda_1 + \lambda_2)$ (C) $\frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$ (D) $\lambda = \sqrt{\lambda_1 \lambda_2}$

13. The radioactive nucleus of an element X decays to a stable nucleus of element Y. A graph of the rate of formation of Y against time would look like



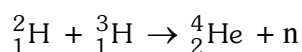
14. The half life of a neutron is 800 sec. 10^8 neutrons at a certain instant are projected from one space station towards another space station, situated 3200 km away, with a velocity 2000 m/s. Their velocity remains constant during the journey. How many neutrons reach the other station?

(A) 50×10^6 (B) 25×10^6 (C) 80×10^5 (D) 25×10^5

15. A radioactive source in the form of a metal sphere of diameter 3.2×10^{-3} m emits β -particle at a constant rate of 6.25×10^{10} particle/sec. The source is electrically insulated and all the β -particle are emitted from the surface. The potential of the sphere will rise to 1 V in time

(A) 180 μ sec (B) 90 μ sec (C) 18 μ sec (D) 9 μ sec

16. In the nuclear fusion reaction,



given that the repulsive potential energy between the two nuclei is 7.7×10^{-14} J, the temperature at which the gases must be heated to initiate the reaction is nearly [Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K]- **[AIEEE - 2003]**

- (A) 10^7 K (B) 10^5 K (C) 10^3 K (D) 10^9 K

17. The binding energy per nucleon of deuteron (${}^2_1\text{H}$) and helium nucleus (${}^4_2\text{He}$) is 1.1 MeV and 7 MeV respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is- **[AIEEE - 2004]**

- (A) 13.9 MeV (B) 26.9 MeV (C) 23.6 MeV (D) 19.2 MeV

MULTIPLE CORRECT TYPE QUESTIONS

18. When a nucleus with atomic number Z and mass number A undergoes a radioactive decay process:

- (A) both Z and A will decrease, if the process is α decay
 (B) Z will decrease but A will not change, if the process is β^+ decay
 (C) Z will decrease but A will not change, if the process is β^- decay
 (D) Z and A will remain unchanged, if the process is γ decay.

19. When the atomic number A of the nucleus increases

- (A) initially the neutron-proton ratio is constant = 1
 (B) initially neutron-proton ratio increases and later decreases
 (C) initially binding energy per nucleon increases and later decreases
 (D) the binding energy per nucleon increases when the neutron-proton ratio increases.

20. Let m_p be the mass of a proton, m_n the mass of a neutron, M_1 the mass of a ${}^{20}_{10}\text{Ne}$ nucleus and M_2 the mass of a ${}^{40}_{20}\text{Ca}$ nucleus. Then

- (A) $M_2 = 2M_1$ (B) $M_2 > 2M_1$ (C) $M_2 < 2M_1$ (D) $M_1 < 10(m_n + m_p)$

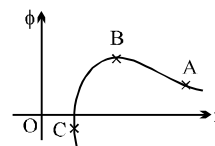
21. The decay constant of a radio active substance is $0.173 \text{ (years)}^{-1}$. Therefore :

- (A) Nearly 63% of the radioactive substance will decay in $(1/0.173)$ year.
 (B) half life of the radio active substance is $(1/0.173)$ year.
 (C) one -forth of the radioactive substance will be left after nearly 8 years.
 (D) all the above statements are true.

22. Which of the following statement(s) is/are correct?

- (A) The rest mass of a stable nucleus is less than the sum of the rest masses of its separated nucleons.
 (B) The rest mass of a stable nucleus is greater than the sum of the rest masses of its separated nucleons.
 (C) In nuclear fusion, energy is released by fusion of two nuclei of medium mass (approx. 100 amu)
 (D) In nuclear fission, energy is released by fragmentation of a very heavy nucleus.

23. The graph shown by the side shows the variation of potential energy ϕ of a proton with its distance 'r' from a fixed sodium nucleus, as it approaches the nucleus, placed at origin O. Then the portion.



- (A) AB indicates nuclear repulsion
(B) AB indicates electrostatic repulsion
(C) BC indicates nuclear attraction
(D) BC represents electrostatic interaction
24. The instability of the nucleus can be due to various causes. An unstable nucleus emits radiations if possible to transform into less unstable state. Then the cause and the result can be
(A) a nucleus of excess nucleons is α active
(B) an excited nucleus of excess protons is β^- active
(C) an excited nucleus of excess protons is β^+ active
(D) an nucleus of excess neutrons is β^- active
25. In β -decay, the Q-value of the process is E. Then
(A) K.E. of a β -particle cannot exceed E.
(B) K.E. of anti neutrino emitted lies between Zero and E.
(C) N/Z ratio of the nucleus is altered.
(D) Mass number (A) of the nucleus is altered.
26. Consider the following nuclear reactions and select the correct statements from the options that follow.

Reaction I : $n \rightarrow p + e^- + \bar{\nu}$

Reaction II : $p \rightarrow n + e^+ + \nu$

- (A) Free neutron is unstable, therefore reaction I is possible
(B) Free proton is stable, therefore reaction II is not possible
(C) Inside a nucleus, both decays (reaction I and II) are possible
(D) Inside a nucleus, reaction I is not possible but reaction II is possible.
27. When the nucleus of an electrically neutral atom undergoes a radioactive decay process, it will remain neutral after the decay if the process is :
(A) α decay (B) β^- decay (C) γ decay (D) K-capture

MATRIX MATCH TYPE QUESTION

28. In the following, column I lists some physical quantities & the column II gives approx. energy values associated with some of them. Choose the appropriate value of energy from column II for each of the physical quantities in column I and write the corresponding letter A, B, C etc. against the number (i), (ii), (iii), etc. of the physical quantity in the answer book. In your answer, the sequence of column I should be maintained.

Column I

- (i) Energy of thermal neutrons
(ii) Energy of X-rays
(iii) Binding energy per nucleon
(iv) Photoelectric threshold of metal

Column II

- (A) 0.025 eV
(B) 0.5 eV
(C) 3 eV
(D) 20 eV
(E) 10 keV
(F) 8 MeV

EXERCISE - JM

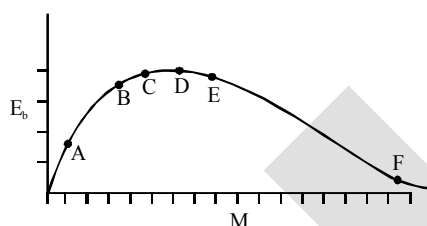
1. This question contains Statement-1 and Statement-2. Out of the four choices given after the statements, choose the one that best describes the two statements.

Statement-1 : Energy is released when heavy nuclei undergo fission or light nuclei undergo fusion.

Statement-2 : For heavy nuclei, binding energy per nucleon increases with increasing Z while for light nuclei it decreases with increasing Z . [AIEEE - 2008]

- (1) Statement-1 is false, Statement-2 is true.
 (2) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
 (3) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.
 (4) Statement-1 is true, Statement-2 is false.

2.



The above is a plot of binding energy per nucleon E_b , against the nuclear mass M ; A, B, C, D, E, F correspond to different nuclei. Consider four reactions : [AIEEE - 2008]

- (i) $A + B \rightarrow C + \varepsilon$ (ii) $C \rightarrow A + B + \varepsilon$
 (iii) $D + E \rightarrow F + \varepsilon$ (iv) $F \rightarrow D + E + \varepsilon$

where ε is the energy released ? In which reactions is ε positive ?

- (i) $A + B \rightarrow C + \varepsilon$ (ii) $C \rightarrow A + B + \varepsilon$ (iii) $D + E \rightarrow F + \varepsilon$ (iv) $F \rightarrow D + E + \varepsilon$
 (1) (ii) and (iv) (2) (ii) and (iii) (3) (i) and (iv) (4) (i) and (iii)

Directions : Questions number 3 – 4 are based on the following paragraph.

A nucleus of mass $M + \Delta m$ is at rest and decays into two daughter nuclei of equal mass $\frac{M}{2}$ each.

Speed of light is c .

[AIEEE-2010]

3. The speed of daughter nuclei is :-

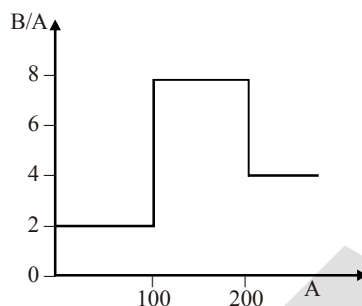
- (1) $c \sqrt{\frac{\Delta m}{M + \Delta m}}$ (2) $c \frac{\Delta m}{M + \Delta m}$ (3) $c \sqrt{\frac{2\Delta m}{M}}$ (4) $c \sqrt{\frac{\Delta m}{M}}$

4. The binding energy per nucleon for the parent nucleus is E_1 and that for the daughter nuclei is E_2 . Then:- [AIEEE - 2010]
 (1) $E_1 = 2E_2$ (2) $E_2 = 2E_1$ (3) $E_1 > E_2$ (4) $E_2 > E_1$
5. A radioactive nucleus (initial mass number A and atomic number Z) emits 3 α -particles and 2 positrons. The ratio of number of neutrons to that of protons in the final nucleus will be:- [AIEEE - 2010]
 (1) $\frac{A-Z-4}{Z-2}$ (2) $\frac{A-Z-8}{Z-4}$ (3) $\frac{A-Z-4}{Z-8}$ (4) $\frac{A-Z-12}{Z-4}$
6. The half life of a radioactive substance is 20 minutes. The approximate time interval ($t_2 - t_1$) between the time t_2 when $\frac{2}{3}$ of it has decayed and time t_1 when $\frac{1}{3}$ of it had decayed is :- [AIEEE - 2011]
 (1) 20 min (2) 28 min (3) 7 min (4) 14 min
7. After absorbing a slowly moving neutron of mass m_N (momentum ~ 0) a nucleus of mass M breaks into two nuclei of masses m_1 and $5m_1$ ($6m_1 = M + m_N$), respectively. If the de Broglie wavelength of the nucleus with mass m_1 is λ , then de Broglie wavelength of the other nucleus will be:- [AIEEE - 2011]
 (1) 25λ (2) 5λ (3) $\frac{\lambda}{5}$ (4) λ
8. **Statement-1:** A nucleus having energy E_1 decays by β^- emission to daughter nucleus having energy E_2 , but the β^- rays are emitted with a continuous energy spectrum having end point energy $E_1 - E_2$.
Statement-2: To conserve energy and momentum in β -decay at least three particles must take part in the transformation. [AIEEE - 2011]
 (1) Statement-1 is incorrect, statement-2 is correct
 (2) Statement-1 is correct, statement-2 is incorrect
 (3) Statement-1 is correct, statement-2 correct; statement-2 is the correct explanation of statement-1
 (4) Statement-1 is correct, statement-2 is correct; statement-2 is not the correct explanation of statement-1.
9. Assume that a neutron breaks into a proton and an electron. The energy released during this process is : [AIEEE - 2012]
 (Mass of neutron = 1.6725×10^{-27} kg)
 Mass of proton = 1.6725×10^{-27} kg
 Mass of electron = 9×10^{-31} kg)
 (1) 5.4 MeV (2) 0.73 MeV (3) 7.10 MeV (4) 6.30 MeV
10. Hydrogen (${}_1H^1$), Deuterium (${}_1H^2$), singly ionised Helium (${}_2He^4$)⁺ and doubly ionised lithium (${}_3Li^6$)⁺⁺ all have one electron around the nucleus. Consider an electron transition from $n = 2$ to $n = 1$. If the wave lengths of emitted radiation are $\lambda_1, \lambda_2, \lambda_3$ and λ_4 respectively then approximately which one of the following is correct ? [JEE-Main-2014]
 (1) $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$ (2) $\lambda_1 = 2\lambda_2 = 3\lambda_3 = 4\lambda_4$
 (3) $4\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$ (4) $\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

11. The radiation corresponding to $3 \rightarrow 2$ transition of hydrogen atom falls on a metal surface to produce photoelectrons. These electrons are made to enter a magnetic field of $3 \times 10^{-4} \text{ T}$. If the radius of the largest circular path followed by these electrons is 10.0 mm, the work function of the metal is close to:-
[JEE-Main-2014]
(1) 0.8 eV (2) 1.6 eV (3) 1.8 eV (4) 1.1 eV
12. Half-lives of two radioactive elements A and B are 20 minutes and 40 minutes, respectively. Initially, the samples have equal number of nuclei. After 80 minutes, the ratio of decayed numbers of A and B nuclei will be :-
[JEE-Main-2016]
(1) 5 : 4 (2) 1 : 16 (3) 4 : 1 (4) 1 : 4
13. A radioactive nucleus A with a half life T , decays into a nucleus B. At $t = 0$, there is no nucleus B. At sometime t , the ratio of the number of B to that of A is 0.3. Then, t is given by :
[JEE-Main-2017]
(1) $t = T \log (1.3)$ (2) $t = \frac{T}{\log(1.3)}$ (3) $t = \frac{T \log 2}{2 \log 1.3}$ (4) $t = T \frac{\log 1.3}{\log 2}$
14. An electron from various excited states of hydrogen atom emit radiation to come to the ground state. Let λ_n, λ_g be the de Broglie wavelength of the electron in the n^{th} state and the ground state respectively. Let Λ_n be the wavelength of the emitted photon in the transition from the n^{th} state to the ground state. For large n , (A, B are constants)
[JEE-Main-2018]
(1) $\Lambda_n \approx A + B\lambda_n$ (2) $\Lambda_n^2 \approx A + B\lambda_n^2$ (3) $\Lambda_n^2 \approx \lambda_n$ (4) $\Lambda_n \approx A + \frac{B}{\lambda_n^2}$

EXERCISE - JA

1. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Use this plot to choose the correct choice(s) given below : **[JEE 2008]**



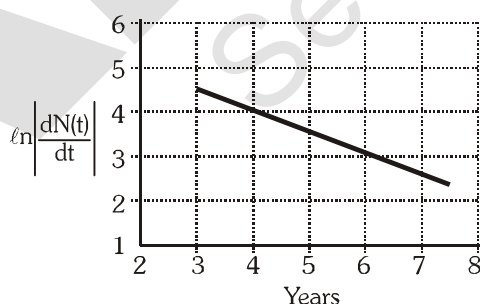
- (A) Fusion of two nuclei with mass numbers lying in the range of $1 < A < 50$ will release energy
- (B) Fusion of two nuclei with mass numbers lying in the range of $51 < A < 100$ will release energy
- (C) Fission of a nucleus lying in the mass range of $100 < A < 200$ will release energy when broken into two equal fragments
- (D) Fission of a nucleus lying in the mass range of $200 < A < 260$ will release energy when broken into two equal fragments
2. A radioactive sample S1 having an activity of $5\mu\text{Ci}$ has twice the number of nuclei as another sample S2 which has an activity of $10\mu\text{Ci}$. The half lives of S1 and S2 can be : **[JEE 2008]**
- (A) 20 years and 5 years, respectively
- (B) 20 years and 10 years, respectively
- (C) 10 years each
- (D) 5 years each

Paragraph for Question Nos. 3 to 5

Scientists are working hard to develop nuclear fusion reactor. Nuclei of heavy hydrogen, ${}^2_1\text{H}$, known as deuteron and denoted by D, can be thought of as a candidate for fusion reactor. The D-D reaction is ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^3_2\text{He} + \text{n} + \text{energy}$. In the core of fusion reactor, a gas of heavy hydrogen is fully ionized into deuteron nuclei and electrons. This collection of ${}^2_1\text{H}$ nuclei and electrons is known as plasma. The nuclei move randomly in the reactor core and occasionally come close enough for nuclear fusion to take place. Usually, the temperatures in the reactor core are too high and no material wall can be used to confine the plasma. Special techniques are used which confine the plasma for a time t_0 before the particles fly away from the core. If n is the density (number/volume) of deuterons, the product nt_0 is called Lawson number. In one of the criteria, a reactor is termed successful if Lawson number is greater than $5 \times 10^{14} \text{ s/cm}^3$. It may be helpful to use the following :

Boltzmann constant $k = 8.6 \times 10^{-5} \text{ eV/K}$; $\frac{e^2}{4\pi\epsilon_0} = 1.44 \times 10^{-9} \text{ eVm}$.

3. In the core of nuclear fusion reactor, the gas becomes plasma because of [JEE-2009]
 (A) strong nuclear force acting between the deuterons
 (B) Coulomb force acting between the deuterons
 (C) Coulomb force acting between deuteron-electron pairs
 (D) the high temperature maintained inside the reactor core
4. Assume that two deuteron nuclei in the core of fusion reactor at temperature T are moving towards each other, each with kinetic energy $1.5 kT$, when the separation between them is large enough to neglect Coulomb potential energy. Also neglect any interaction from other particles in the core. The minimum temperature T required for them to reach a separation of $4 \times 10^{-15} \text{ m}$ is in the range [JEE-2009]
 (A) $1.0 \times 10^9 \text{ K} < T < 2.0 \times 10^9 \text{ K}$ (B) $2.0 \times 10^9 \text{ K} < T < 3.0 \times 10^9 \text{ K}$
 (C) $3.0 \times 10^9 \text{ K} < T < 4.0 \times 10^9 \text{ K}$ (D) $4.0 \times 10^9 \text{ K} < T < 5.0 \times 10^9 \text{ K}$
5. Results of calculations for four different designs of a fusion reactor using D-D reaction are given below. Which of these is most promising based on Lawson criterion? [JEE-2009]
 (A) deuteron density $= 2.0 \times 10^{12} \text{ cm}^{-3}$, confinement time $= 5.0 \times 10^{-3} \text{ s}$
 (B) deuteron density $= 8.0 \times 10^{14} \text{ cm}^{-3}$, confinement time $= 9.0 \times 10^{-1} \text{ s}$
 (C) deuteron density $= 4.0 \times 10^{23} \text{ cm}^{-3}$, confinement time $= 1.0 \times 10^{-11} \text{ s}$
 (D) deuteron density $= 1.0 \times 10^{24} \text{ cm}^{-3}$, confinement time $= 4.0 \times 10^{-12} \text{ s}$
6. To determine the half life of a radioactive element, a student plots a graph of $\ln \left| \frac{dN(t)}{dt} \right|$ versus t . Here $\frac{dN(t)}{dt}$ is the rate of radioactive decay at time t . If the number of radioactive nuclei of this element decreases by a factor of p after 4.16 years, the value of p is [JEE-2010]



7. The activity of a freshly prepared radioactive sample is 10^{10} disintegrations per second, whose mean life is 10^9 s . The mass of an atom of this radioisotope is 10^{-25} kg . The mass (in mg) of the radioactive sample is [JEE-2011]

Paragraph for Questions 8 and 9

The β -decay process, discovered around 1900, is basically the decay of a neutron (n). In the laboratory, a proton (p) and an electron (e^-) are observed as the decay products of the neutron. Therefore, considering the decay of a neutron as a two-body decay process, it was predicted theoretically that the kinetic energy of the electron should be a constant. But experimentally, it was observed that the electron kinetic energy has a continuous spectrum. Considering a three-body decay process, i.e. $n \rightarrow p + e^- + \bar{\nu}_e$, around 1930, Pauli explained the observed electron energy spectrum. Assuming the anti-neutrino ($\bar{\nu}_e$) to be massless and possessing negligible energy, and the neutron to be at rest, momentum and energy conservation principles are applied. From this calculation, the maximum kinetic energy of the electron is 0.8×10^6 eV. The kinetic energy carried by the proton is only the recoil energy.

8. If the anti-neutrino had a mass of $3 \text{ eV}/c^2$ (where c is the speed of light) instead of zero mass, what should be the range of the kinetic energy, K , of the electron? [JEE 2012]
- (A) $0 \leq K \leq 0.8 \times 10^6 \text{ eV}$ (B) $3.0 \text{ eV} \leq K \leq 0.8 \times 10^6 \text{ eV}$
 (C) $3.0 \text{ eV} \leq K < 0.8 \times 10^6 \text{ eV}$ (D) $0 \leq K < 0.8 \times 10^6 \text{ eV}$
9. What is the maximum energy of the anti-neutrino? [JEE 2012]
- (A) zero (B) much less than $0.8 \times 10^6 \text{ eV}$
 (C) Nearly $0.8 \times 10^6 \text{ eV}$ (D) Much larger than $0.8 \times 10^6 \text{ eV}$
10. A freshly prepared sample of a radioisotope of half-life 1386 s has activity 10^3 disintegrations per second. Given that $\ln 2 = 0.693$, the fraction of the initial number of nuclei (expressed in nearest integer percentage) that will decay in the first 80 s after preparation of the sample is. [JEE Advance-2013]

Paragraph for Questions 11 and 12

The mass of a nucleus ${}_Z^AX$ is less than the sum of the masses of $(A - Z)$ number of neutrons and Z number of protons in the nucleus. The energy equivalent to the corresponding mass difference is known as the binding energy of the nucleus. A heavy nucleus of mass M can break into two light nuclei of masses m_1 and m_2 only if $(m_1 + m_2) < M$. Also two light nuclei of masses m_3 and m_4 can undergo complete fusion and form a heavy nucleus of mass M' only if $(m_3 + m_4) > M'$. The masses of some neutral atoms are given in the table below :- [JEE Advance-2013]

${}_1^1\text{H}$	1.007825 u	${}_1^2\text{H}$	2.014102 u	${}_1^3\text{H}$	3.016050 u	${}_2^4\text{He}$	4.002603 u
${}_3^6\text{Li}$	6.015123 u	${}_3^7\text{Li}$	7.016004 u	${}_{30}^{70}\text{Zn}$	69.925325 u	${}_{34}^{82}\text{Se}$	81.916709 u
${}_{64}^{152}\text{Gd}$	151.919803 u	${}_{82}^{206}\text{Pb}$	205.974455 u	${}_{83}^{209}\text{Bi}$	208.980388 u	${}_{84}^{210}\text{Po}$	209.982876 u

$(1\text{u} = 932 \text{ MeV}/c^2)$

11. The kinetic energy (in keV) of the alpha particle, when the nucleus ${}_{84}^{210}\text{Po}$ at rest undergoes alpha decay, is :-
- (A) 5319 (B) 5422 (C) 5707 (D) 5818

12. The correct statement is :-

- (A) The nucleus ${}^6_3\text{Li}$ can emit an alpha particle
 (B) The nucleus ${}^{210}_{84}\text{Po}$ can emit a proton
 (C) Deuteron and alpha particle can undergo complete fusion
 (D) The nuclei ${}^{70}_{30}\text{Zn}$ and ${}^{82}_{34}\text{Se}$ can undergo complete fusion

13. Match List I of the nuclear processes with List II containing parent nucleus and one of the end products of each process and then select the correct answer using the codes given below the lists:

[JEE Advance-2013]

List I	List II
P. Alpha decay	1. ${}^{15}_8\text{O} \rightarrow {}^{15}_7\text{N} + \dots$
Q. β^+ decay	2. ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + \dots$
R. Fission	3. ${}^{185}_{83}\text{Bi} \rightarrow {}^{184}_{82}\text{Pb} + \dots$
S. Proton emission	4. ${}^{239}_{94}\text{Pu} \rightarrow {}^{140}_{57}\text{La} + \dots$

Codes :

	P	Q	R	S
(A)	4	2	1	3
(B)	1	3	2	4
(C)	2	1	4	3
(D)	4	3	2	1

14. If λ_{Cu} is the wavelength of K_α X-ray line of copper (atomic number 29) and λ_{Mo} is the wavelength

of the K_α X-ray line of molybdenum (atomic number 42), then the ratio $\frac{\lambda_{\text{Cu}}}{\lambda_{\text{Mo}}}$ is close to :-

[JEE Advance-2014]

- (A) 1.99 (B) 2.14 (C) 0.50 (D) 0.48

15. A metal surface is illuminated by light of two different wavelength 248 nm and 310 nm. The maximum speeds of the photoelectrons corresponding to these wavelengths are u_1 and u_2 , respectively. If the ratio $u_1 : u_2 = 2 : 1$ and $hc = 1240 \text{ eV nm}$, the work function of the metal is nearly :-

[JEE Advance-2014]

- (A) 3.7 eV (B) 3.2 eV (C) 2.8 eV (D) 2.5 eV

16. A nuclear power plant supplying electrical power to a village uses a radioactive material of half life T years as the fuel. The amount of fuel at the beginning is such that the total power requirement of the village is 12.5% of the electrical power available from the plant at that time. If the plant is able to meet the total power needs of the village for a maximum period of nT years, then the value of n is.

[JEE Advance-2015]

17. Match the nuclear processes given in column-I with the appropriate option(s) in column-II.

[JEE Advance-2015]

Column-I

Column-II

- | | |
|----------------------------------|--|
| (A) Nuclear fusion | (P) Absorption of thermal neutrons by ${}_{92}^{235}\text{U}$ |
| (B) Fission in a nuclear reactor | (Q) ${}_{27}^{60}\text{Co}$ nucleus |
| (C) β -decay | (R) Energy production in stars via hydrogen conversion to helium |
| (D) γ -ray emission | (S) Heavy water |
| | (T) Neutrino emission |

18. For a radioactive material, its activity A and rate of change of its activity R are defined as

$A = -\frac{dN}{dt}$ and $R = -\frac{dA}{dt}$, where $N(t)$ is the number of nuclei at time t . Two radioactive sources P (mean life τ) and Q (mean life 2τ) have the same activity at $t = 0$. Their rates of change of activities at $t = 2\tau$ are R_P and R_Q , respectively. If $\frac{R_P}{R_Q} = \frac{n}{e}$, then the value of n is :

[JEE Advance-2015]

19. A fission reaction is given by ${}_{92}^{236}\text{U} \rightarrow {}_{54}^{140}\text{Xe} + {}_{38}^{94}\text{Sr} + x + y$, where x and y are two particles.

Considering ${}_{92}^{236}\text{U}$ to be at rest, the kinetic energies of the products are denoted by $K_{\text{Xe}}, K_{\text{Sr}}, K_x$ (2 MeV) and K_y (2 MeV), respectively. Let the binding energies per nucleon of ${}_{92}^{236}\text{U}$, ${}_{54}^{140}\text{Xe}$ and ${}_{38}^{94}\text{Sr}$ be 7.5 MeV, 8.5 MeV and 8.5 MeV, respectively. Considering different conservation laws, the correct option(s) is (are) :-

[JEE Advance-2015]

- (A) $x = n, y = n, K_{\text{Sr}} = 129 \text{ MeV}, K_{\text{Xe}} = 86 \text{ MeV}$
 (B) $x = p, y = e^-, K_{\text{Sr}} = 129 \text{ MeV}, K_{\text{Xe}} = 86 \text{ MeV}$
 (C) $x = p, y = n, K_{\text{Sr}} = 129 \text{ MeV}, K_{\text{Xe}} = 86 \text{ MeV}$
 (D) $x = n, y = n, K_{\text{Sr}} = 86 \text{ MeV}, K_{\text{Xe}} = 129 \text{ MeV}$

20. The isotope ${}_{5}^{12}\text{B}$ having a mass 12.014 u undergoes β -decay to ${}_{6}^{12}\text{C}$. ${}_{6}^{12}\text{C}$ has an excited state of the nucleus (${}_{6}^{12}\text{C}^*$) at 4.041 MeV above its ground state. If ${}_{5}^{12}\text{B}$ decays to ${}_{6}^{12}\text{C}^*$, the maximum kinetic energy of the β -particle in units of MeV is ($1u = 931.5 \text{ MeV}/c^2$, where c is the speed of light in vacuum).

[JEE Advance-2016]

21. The electrostatic energy of Z protons uniformly distributed throughout a spherical nucleus of

radius R is given by $E = \frac{3}{5} \frac{Z(Z-1)e^2}{4\pi\epsilon_0 R}$

[JEE Advance-2016]

The measured masses of the neutron, ${}_{1}^1\text{H}$, ${}_{7}^{15}\text{N}$ and ${}_{8}^{15}\text{O}$ are 1.008665 u, 1.007825 u, 15.000109 u and 15.003065 u respectively. Given that the radii of both the ${}_{7}^{15}\text{N}$ and ${}_{8}^{15}\text{O}$ nuclei are same, $1u = 931.5$

MeV/c^2 (c is the speed of light) and $\frac{e^2}{(4\pi\epsilon_0)} = 1.44 \text{ MeV fm}$. Assuming that the difference between

the binding energies of ${}_{7}^{15}\text{N}$ and ${}_{8}^{15}\text{O}$ is purely due to the electrostatic energy, the radius of either of the nuclei is ($1\text{fm} = 10^{-15}\text{m}$)

- (A) 2.85 fm (B) 3.03 fm (C) 3.42 fm (D) 3.80 fm

22. An accident in a nuclear laboratory resulted in deposition of a certain amount of radioactive material of half-life 18 days inside the laboratory. Tests revealed that the radiation was 64 times more than the permissible level required for safe operation of the laboratory. What is the minimum number of days after which the laboratory can be considered safe for use? [JEE Advance-2016]
(A) 64 (B) 90 (C) 108 (D) 120
23. ^{131}I is an isotope of Iodine that β decays to an isotope of Xenon with a half-life of 8 days. A small amount of a serum labelled with ^{131}I is injected into the blood of a person. The activity of the amount of ^{131}I injected was 2.4×10^5 Becquerel (Bq). It is known that the injected serum will get distributed uniformly in the blood stream in less than half an hour. After 11.5 hours, 2.5 ml of blood is drawn from the person's body, and gives an activity of 115 Bq. The total volume of blood in the person's body, in liters is approximately (you may use $e^x \approx 1 + x$ for $|x| \ll 1$ and $\ln 2 \approx 0.7$). [JEE Advance-2017]
24. In a radioactive decay chain, $^{232}_{90}\text{Th}$ nucleus decays to $^{212}_{82}\text{Pb}$ nucleus. Let N_α and N_β be the number of α and β^- particles, respectively, emitted in this decay process. Which of the following statements is (are) true ? [JEE Advance-2018]
(A) $N_\alpha = 5$ (B) $N_\alpha = 6$ (C) $N_\beta = 2$ (D) $N_\beta = 4$

ANSWER KEY

EXERCISE (S-1)

1. Ans. $8/3 \times 10^{18}$ sec 2. Ans. 28 MeV 3. Ans. $v = 1.546 \times 10^{18}$ Hz; $Z = 26$

4. Ans. 4.87 MeV 5. Ans. 6.04×10^9 yrs 6. Ans. $t = \left(\frac{\ln 5}{\ln 2} \right) \tau$

EXERCISE (S-2)

1. Ans. 1.7×10^{10} years 2. Ans. 5196 yrs 3. Ans. $v = u\lambda t$

4. Ans. $\Delta T = \frac{0.2E_0 \left[\alpha t - \frac{\alpha}{\lambda} (1 - e^{-\lambda t}) \right]}{mS}$ 5. Ans. $\cong 33.298 \mu W$

6. Ans. 6 litre 7. Ans. 2.4×10^{-12} m 8. Ans. 9.00×10^6

9. Ans. ($T_{1/2} = 10.8$ sec) 10. Ans. $1.75n = N_0(1 - e^{-4\lambda})$, 6.95 sec, $\frac{2}{\ln\left(\frac{4}{3}\right)}$

11. Ans. 0.259

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

- | | | | | | |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (D) | 2. Ans. (C) | 3. Ans. (A) | 4. Ans. (A) | 5. Ans. (D) | 6. Ans. (A) |
| 7. Ans. (B) | 8. Ans. (A) | 9. Ans. (B) | 10. Ans. (D) | 11. Ans. (B) | 12. Ans. (B) |
| 13. Ans. (C) | 14. Ans. (C) | 15. Ans. (A) | 16. Ans. (B) | 17. Ans. (C) | 18. Ans. (B) |
| 19. Ans. (B) | 20. Ans. (B) | 21. Ans. (A) | 22. Ans. (D) | 23. Ans. (D) | 24. Ans. (C) |
| 25. Ans. (B) | 26. Ans. (B) | 27. Ans. (B) | 28. Ans. (C) | 29. Ans. (A) | 30. Ans. (C) |
| 31. Ans. (C) | 32. Ans. (C) | 33. Ans. (A) | 34. Ans. (B) | 35. Ans. (C) | 36. Ans. (C) |
| 37. Ans. (B) | 38. Ans. (B) | 39. Ans. (C) | | | |

MATRIX MATCH TYPE QUESTION

40. Ans. (A) P, Q; (B) P, R; (C) S, P; (D) P, Q, R

41. Ans. (A) R, P; (B) Q, S; (C) P; (D) Q

EXERCISE (O-2)

- | | | | |
|---------------------------------------|--------------------|-----------------|--------------------|
| 1. Ans. (B) | 2. Ans. (B) | 3. Ans. (C) | 4. Ans. (D) |
| 5. Ans. (C) | 6. Ans. (C) | 7. Ans. (B) | 8. Ans. (B) |
| 9. Ans. (C) | 10. Ans. (B) | 11. Ans. (C) | 12. Ans. (A) |
| 13. Ans. (E) | 14. Ans. (B) | 15. Ans. (C) | 16. Ans. (D) |
| 17. Ans. (C) | 18. Ans. (A, B, D) | 19. Ans. (A, C) | 20. Ans. (C, D) |
| 21. Ans. (A, C) | 22. Ans. (A, D) | 23. Ans. (B, C) | 24. Ans. (A, C, D) |
| 25. Ans. (A, B, C) | 26. Ans. (A, B, C) | 27. Ans. (C, D) | |
| 28. Ans. (i) A, (ii) E (iii) F (iv) C | | | |

EXERCISE - JM

- | | | | |
|-----------------|--------------|--------------|--------------|
| 1. Ans. (4) | 2. Ans. (3) | 3. Ans. (3) | 4. Ans. (4) |
| 5. Ans. (3) | 6. Ans. (1) | 7. Ans. (4) | 8. Ans. (3) |
| 9. Ans. (Bonus) | 10. Ans. (1) | 11. Ans. (4) | 12. Ans. (1) |
| 13. Ans. (4) | 14. Ans. (4) | | |

EXERCISE - JA

- | | | | |
|--|----------------|--------------|--------------|
| 1. Ans. (B, D) | 2. Ans. (A) | 3. Ans. (D) | 4. Ans. (A) |
| 5. Ans. (B) | 6. Ans. (8) | 7. Ans. (1) | 8. Ans. (D) |
| 9. Ans. (C) | 10. Ans. 4 | 11. Ans. (A) | 12. Ans. (C) |
| 13. Ans. (C) | 14. Ans. (B) | 15. Ans. (A) | 16. Ans. 3 |
| 17. Ans. (A)-R or R,T; (B)-P & S; (C)-Q & T; (D)-R | | | 18. Ans. 2 |
| 19. Ans. (A) | 20. Ans. 9 | 21. Ans. (C) | 22. Ans. (C) |
| 23. Ans. 5 | 24. Ans. (A,C) | | |